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Roll Number: 13

Semester: 6th

Subject: Artificial Neural Networks

Programme: Bachelor of Science Computer Science

Shift: Morning

Session: 2023-2027

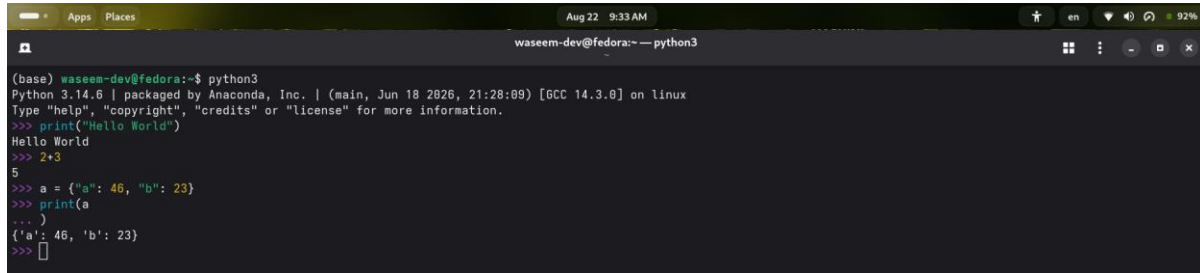
Submitted To: Dr. Abdul Basit

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Artificial Neural Networks

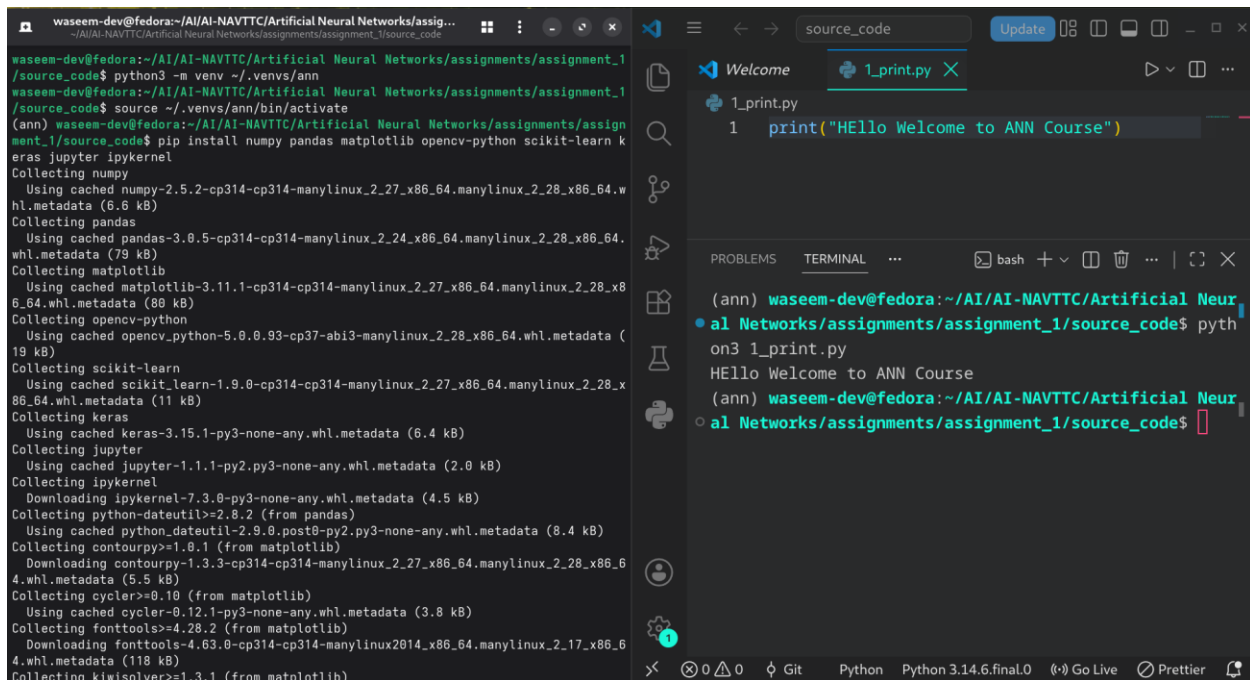
Assignment 1

1) Downloaded Fedora 44 and Python installed:



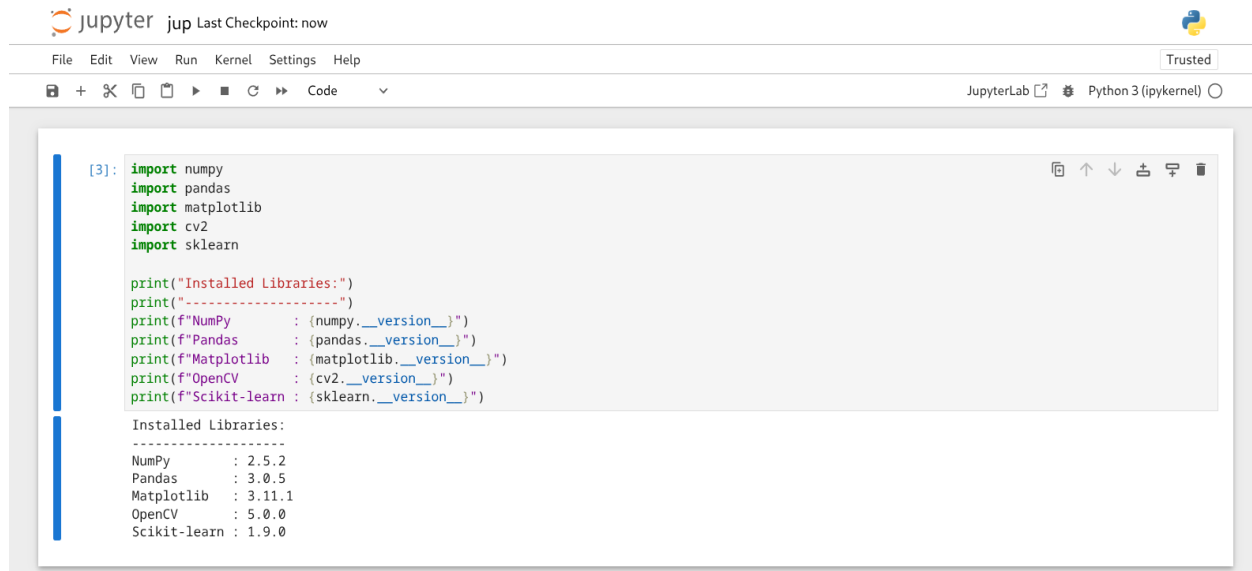
```
(base) waseem-dev@fedora:~$ python3
Python 3.14.6 | packaged by Anaconda, Inc. | (main, Jun 18 2026, 21:28:09) [GCC 14.3.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> print("Hello World")
Hello World
>>> 2+3
5
>>> a = {"a": 46, "b": 23}
>>> print(a)
{'a': 46, 'b': 23}
>>>
```

2) Created *ann* Environment and Installed VS Code



```
waseem-dev@fedora:~/AI/AI-NAVTT/Artificial Neural Networks/assignments/assignment_1/source_code$ python3 -m venv ~/.venvs/ann
waseem-dev@fedora:~/AI/AI-NAVTT/Artificial Neural Networks/assignments/assignment_1/source_code$ source ~/.venvs/ann/bin/activate
(ann) waseem-dev@fedora:~/AI/AI-NAVTT/Artificial Neural Networks/assignments/assignment_1/source_code$ pip install numpy pandas matplotlib opencv-python scikit-learn keras jupyter ipykernel
Collecting numpy
  Using cached numpy-2.5.2-cp314-cp314-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Collecting pandas
  Using cached pandas-3.0.5-cp314-cp314-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)
Collecting matplotlib
  Using cached matplotlib-3.11.1-cp314-cp314-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (80 kB)
Collecting opencv-python
  Using cached opencv-python-5.0.0.93-cp37-ab13-manylinux_2_28_x86_64.whl.metadata (19 kB)
Collecting scikit-learn
  Using cached scikit_learn-1.9.0-cp314-cp314-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
Collecting keras
  Using cached keras-3.15.1-py3-none-any.whl.metadata (6.4 kB)
Collecting jupyter
  Using cached jupyter-1.1.1-py2.py3-none-any.whl.metadata (2.0 kB)
Collecting ipykernel
  Downloading ipykernel-7.3.0-py3-none-any.whl.metadata (4.5 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.3-cp314-cp314-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.63.0-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (118 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
```

3) Installed Jupyter Notebook in my *ann* environment



The screenshot shows a JupyterLab interface with a code cell containing the following Python code:

```
[3]: import numpy
import pandas
import matplotlib
import cv2
import sklearn

print("Installed Libraries:")
print("-----")
print(f"NumPy      : {numpy.__version__}")
print(f"Pandas     : {pandas.__version__}")
print(f"Matplotlib  : {matplotlib.__version__}")
print(f"OpenCV      : {cv2.__version__}")
print(f"Scikit-learn : {sklearn.__version__}")
```

The output of the code cell is:

```
Installed Libraries:
-----
NumPy      : 2.5.2
Pandas     : 3.0.5
Matplotlib : 3.11.1
OpenCV     : 5.0.0
Scikit-learn : 1.9.0
```